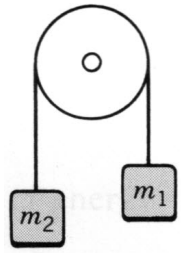
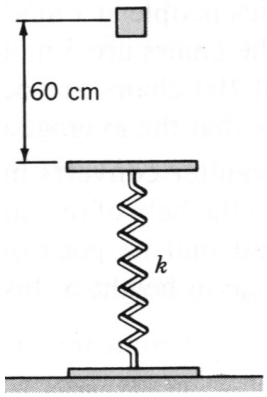
Homework #10

1) Two blocks of masses $m_1 = 5.00$ kg and $m_2 = 2.00$ kg hang on either side of a frictionless cylinder as the figure below. If the system starts at rest, what is the speed of m_1 after it has fallen 40.0 cm?



2) The metabolism of 1.0 g of fat releases 3.76 x 10^4 J. A 70.0 kg person walks to the top of the 433 m Sears tower. Assuming a 15% efficiency in converting the released chemical energy to mechanical energy, how much mass could the person lose?

3) A 500-g block is dropped from a height of 60.0 cm above the top of a vertical spring whose stiffness constant is $k = 120$ N/m as shown in the figure below. Find the maximum compression. (You will need to solve a quadratic equation.)



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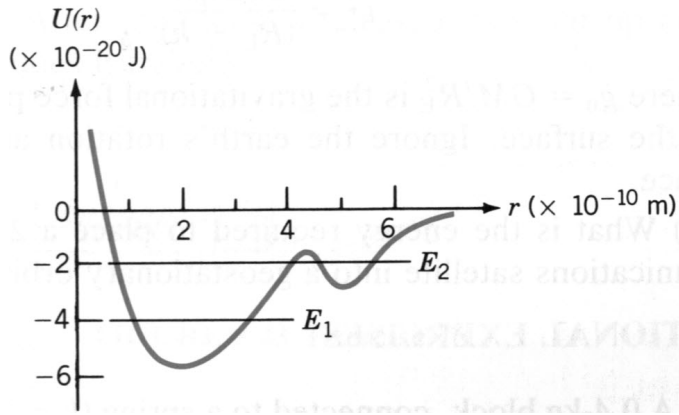
4) Consider the potential energy function below. What are the turning points if the energy of the particle is (a) E_1or (b)E_2? (c) What is the kinetic energy of the particle at position 2.0×10^{-10} m when its total energy is E_1?



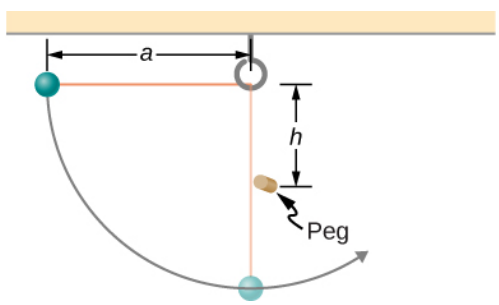
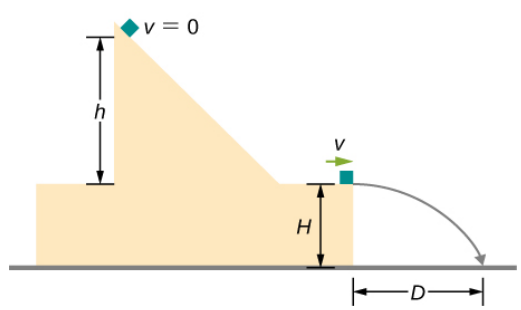
5) Find the potential energy function for the force $F(x) = \frac{b}{x^2}$. Take $U = 0$ at $x = \infty$.

6) The Yukawa potential energy function for the interactions of neutrons and protons in a nucleus is

$$U(r) = -\left(\frac{r_0}{r}\right) U_0 e^{-(r/r_0)}$$

(a) What is the force function $F(r)$? (b) Compute the value of the force at $r = r_0$, and $r = 3r_0$. Take $U_0 = 5.0 \times 10^{-12}$ J and $r_0 = 1.5 \times 10^{-15}$ m.

7) Shown below is a small ball of mass m attached to a string of length a. A small peg is located a distance h below the point where the string is supported. If the ball is released when the string is horizontal, show that h must be greater than $3a/5$ if the ball is to swing completely around the peg.



8) A block leaves a frictionless inclined surface horizontally after dropping off by a height h. Find the horizontal distance D where it will land on the floor, in terms of h, H, and g.